

Database Design Document

Train Reservation System..

WEEK 2: ENTITY, ATTRIBUTES AND ENTITY RELATIONSHIP DIAGRAM.

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Project Title

Train Reservation System

✓ Entity identification

1. Passenger

Primary Key (PK): Passenger_ID

Attributes:

- Passenger_ID
- Passenger_Name
- Age
- Gender
- Phone_Number
- Email
- Address
- Password

2. Train

Primary Key (PK): Train_ID

Attributes:

- Train_ID
- Train_Name

- Train_Number
- Source_Station
- Destination_Station
- Departure_Time
- Arrival_Time

3. Station

Primary Key (PK): Station_ID

Attributes:

- Station_ID
- Station_Name
- City
- State

4. Reservation

Primary Key (PK): Reservation_ID

Foreign Keys (FK):

- Passenger_ID
- Train_ID

Attributes:

- Reservation_ID
- Booking_Date
- Journey_Date
- Class_Type
- Booking_Status

5. Ticket

Primary Key (PK): Ticket_ID

Foreign Key (FK):

- Reservation_ID

Attributes:

- Ticket_ID
- Seat_Number
- Coach_Number
- Fare
- Ticket_Status

6. Payment

Primary Key (PK): Payment_ID

Foreign Key (FK):

- Reservation_ID

Attributes:

- Payment_ID
- Payment_Method
- Payment_Date
- Amount
- Payment_Status

7. Seat

Primary Key (PK): Seat_ID

Foreign Key (FK):

- Train_ID

Attributes:

- Seat_ID
- Coach_Number
- Seat_Number

- Seat_Type
- Seat_Status

8. Admin

Primary Key (PK): Admin_ID

Attributes:

- Admin_ID
- Admin_Name
- Username
- Password
- Email

✓ Relationships

Relationship Cardinality

Passenger → Reservation	One Passenger can book Many Reservations	(1:M)
Train → Reservation	One Train can have Many Reservations	(1:M)
Reservation → Ticket	One Reservation generates One Ticket	(1:1)
Reservation → Payment	One Reservation has One Payment	(1:1)
Train → Seat	One Train contains Many Seats	(1:M)
Admin → Train	One Admin manages Many Trains	(1:M)
Admin → Reservation	One Admin manages Many Reservations	(1:M)
Station → Train	One Station is connected to Many Trains	(1:M)

✓ Entity-Relationship Model:

TRAIN RESERVATION SYSTEM

ER DIAGRAM

